Calculation questions – 4.1 Energy

AQA GCSE Physics (8463) Higher · Paper 1 · 16 questions · 53 marks · answers at the back

Equations for this topic

Equation	Symbols and units	Exam
Ek = ½mv²	Ek (J), m (kg), v (m/s)	Learn it
Ep = mgh	Ep (J), m (kg), g (N/kg), h (m)	Learn it
Ee = ½ke²	Ee (J), k (N/m), e (m)	On sheet
ΔE = mcΔθ	ΔE (J), m (kg), c (J/kg °C), Δθ (°C)	On sheet
P = E/t and P = W/t	P (W), E or W (J), t (s)	Learn it
E = Pt	E (J), P (W), t (s)	Learn it
efficiency = useful output energy ÷ total input energy	no unit (× 100 for %)	Learn it
efficiency = useful power output ÷ total power input	no unit (× 100 for %)	Learn it
W = Fs and W = mg	work (J), F (N), s (m); weight (N)	Learn it

Write the equation, substitute, then rearrange. Convert to base units first. Give a unit and 2–3 significant figures. Take g = 9.8 N/kg.

Grade 4–5

1 A 2.0 kg ball moves at 3.0 m/s. Calculate its kinetic energy. [2 marks]

Answer: _____ Unit: _____

2 A 50 kg climber climbs 12 m up a rock face. Calculate the gravitational potential energy she gains. (g = 9.8 N/kg) [2 marks]

Answer: _____ Unit: _____

3 A kettle transfers 180 000 J of energy in 60 s. Calculate its power. [2 marks]

Answer: _____ Unit: _____

4 A motor is supplied with 500 J and transfers 350 J usefully. Calculate its efficiency as a percentage. [2 marks]

Answer: _____ Unit: _____

5 A spring with spring constant 40 N/m is stretched by 0.10 m. Calculate the elastic potential energy stored. [2 marks]

Answer: _____ Unit: _____

Grade 6–7

6 A 0.80 kg aluminium block is heated from 20 °C to 45 °C. Specific heat capacity of aluminium = 900 J/kg °C. Calculate the energy transferred. [2 marks]

Answer: _____ Unit: _____

7 A 1200 kg car has 150 kJ of kinetic energy. Calculate its speed. [3 marks]

Answer: _____ Unit: _____

8 A 2.0 kW heater is switched on for 15 minutes. Calculate the energy transferred in joules. Give your answer in standard form. [3 marks]

Answer: _____ Unit: _____

9 A 60 W lamp has an efficiency of 15%. (a) Calculate the useful power output. (b) Calculate the energy wasted in 1 minute. [4 marks]

Answer: _____ Unit: _____

10 2.5 kg of water absorbs 52.5 kJ and its temperature rises by 5.0 °C. Calculate the specific heat capacity of water. [3 marks]

Answer: _____ Unit: _____

11 A 70 kg student does 4900 J of work running up some stairs in 8.0 s. (a) Calculate her power. (b) Calculate the height of the stairs. [4 marks]

Answer: _____ Unit: _____

Grade 8–9

12 A 0.050 kg ball is dropped from 3.2 m. (a) Calculate its speed just before it hits the ground, ignoring air resistance. (b) Explain why you did not need the mass. (c) It rebounds to 2.0 m. Calculate the percentage of its energy dissipated. [6 marks]

Answer: _____ Unit: _____

13 A catapult with spring constant 250 N/m is stretched by 8.0 cm and fires a 20 g stone straight up. Assuming all the elastic energy becomes gravitational potential energy, calculate the maximum height. [4 marks]

Answer: _____ Unit: _____

14 A 2.4 kW kettle heats 1.5 kg of water from 15 °C to 100 °C in 4 minutes 30 seconds. c of water = 4200 J/kg °C. Calculate the efficiency of the kettle. [5 marks]

Answer: _____ Unit: _____

15 In a hydroelectric station, 500 kg of water falls 120 m every second. The generators are 85% efficient. Calculate the electrical power output in kW. [4 marks]

Answer: _____ Unit: _____

16 A 900 kg car accelerates from 10 m/s to 25 m/s. The engine's useful power output is 60 kW. Calculate the minimum time taken. [5 marks]

Answer: _____ Unit: _____

Answers – 4.1 Energy

Mark yourself honestly: a mark for the right equation, a mark for substitution, a mark for the answer with its unit.

Q	Working	Answer
1	$E_k = \frac{1}{2} \times 2.0 \times 3.0^2$	**9.0 J**
2	$E_p = 50 \times 9.8 \times 12$	**5880 J (5900 J)**
3	$P = 180\,000 \div 60$	**3000 W**
4	$350 \div 500 = 0.70$	**70%**
5	$E_e = \frac{1}{2} \times 40 \times 0.10^2$	**0.20 J**
6	$\Delta E = 0.80 \times 900 \times (45 - 20)$	**18 000 J**
7	$150\,000 = \frac{1}{2} \times 1200 \times v^2 \rightarrow v^2 = 250$	**15.8 m/s**
8	$E = 2000\ W \times 900\ s$	**1.8×10^6 J**
9	(a) 0.15×60 (b) wasted power = 51 W; 51×60	**(a) 9.0 W (b) 3060 J**
10	$c = 52\,500 \div (2.5 \times 5.0)$	**4200 J/kg °C**
11	(a) $4900 \div 8.0$ (b) $h = 4900 \div (70 \times 9.8)$	**(a) 610 W (b) 7.1 m**
12	(a) $mgh = \frac{1}{2}mv^2 \rightarrow v = \sqrt{(2 \times 9.8 \times 3.2)}$ (b) m is on both sides so cancels (c) $E_p \propto h$: $(3.2 - 2.0) \div 3.2$	**(a) 7.9 m/s (c) 37.5%**
13	$E_e = \frac{1}{2} \times 250 \times 0.080^2 = 0.80$ J; $h = 0.80 \div (0.020 \times 9.8)$	**4.1 m**
14	useful $= 1.5 \times 4200 \times 85 = 535\,500$ J; input $= 2400 \times 270 = 648\,000$ J; $535\,500 \div 648\,000$	**0.83 (83%)**
15	E_p per second $= 500 \times 9.8 \times 120 = 588\,000$ W; $\times 0.85 = 499\,800$ W	**500 kW**
16	$\Delta E_k = \frac{1}{2} \times 900 \times (25^2 - 10^2) = 236\,250$ J; $t = 236\,250 \div 60\,000$	**3.9 s**

Common mistakes. Squaring the change in speed instead of finding each Ek separately (Q16); not converting minutes to seconds (Q8, Q14) or cm and g to m and kg (Q13).